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Introduction

In this lab, we explore essential Linux administration tasks focused on handling and analyzing log files for security and system monitoring purposes. Understanding log files is a critical aspect of managing network environments, as they provide insight into system events, service activity, and potential security threats. This exercise involves setting up an Ubuntu virtual machine, installing and configuring NGINX, and using various tools such as tail to monitor logs in real-time. These activities help develop the skills necessary to detect and prevent vulnerabilities within a network system. By the end of this lab, students will be able to interpret and manipulate log files effectively, which is a key component of system administration and cybersecurity.

	NAME		
	REGISTRATION NO		MARKS
	DATE		/ 100
	CLASS		
	CODE/SUBJECT	DFS40083-COMPUTER NETWORK SECURITY	
	PRACTICAL TASK	1	
	LECTURER		

CLO 1 : Follow security procedures to prevent vulnerabilities and attacks in a network environment (P3, PLO 3)

TOPIC : 2.4 Display Basic Linux Administration Tasks

DURATION : 3 Hours

Introduction

In this lab, you will get familiar with locating and manipulating Linux log files.

Required Resources

- Ubuntu 20.++ Virtual Machine

Part 1 : Install Ubuntu 20.++ VM ISO

The desktop image allows you to try Ubuntu without changing your computer at all, and at your option to install it permanently later. This type of image is what most people will want to use. You will need at least 1024MiB of RAM to install from this image

Part 2 : Setup nginx

NGINX is open source software for web serving, reverse proxying, caching, load balancing, media streaming, and more. It started out as a web server designed for maximum performance and stability. In addition to its HTTP server capabilities, NGINX can also function as a proxy server for email (IMAP, POP3, and SMTP) and a reverse proxy and load balancer for HTTP, TCP, and UDP servers.

Part 3 : Log File Overview

Log files (also spelled logfiles), are files used by computers to log events. Software programs, background processes, services, or transactions between services, including the operating system itself, may generate such events. Log files are dependent on the application that generates them. It is up to the application developer to conform to log file convention. Software documentation should include information on its log files.

Part 4 : Monitoring Log files in Real Time

As seen in the previous sections, log files can be displayed with many text-presentation tools. While **cat**, **more**, **less**, and **nano** can be used to work with log files, they are not suitable for log file real-time monitoring. Developers designed various tools that allow for log file real-time monitoring. Some tools are text-based while others have a graphical interface. This lab focuses on **tail**, a simple but efficient tool, available in practically every Unix-based system.

Part 5 : Monitoring Log files in Real Time

Change machine settings – network to bridged adapter then Create the new folder in /html with your name and touch example.html then with your friend laptop try to open web browser your machine and try to tail and see the log entry you get, describe what happened

Note: You may be required to create a report as part of this assessment. Your response should be aligning with the provided rubric. Please ensure the report encompasses the following elements:

- a) Cover Page
- b) Introduction
- c) Report must include step by step with explanation
- d) Conclusion
- e) Reference (APA format)

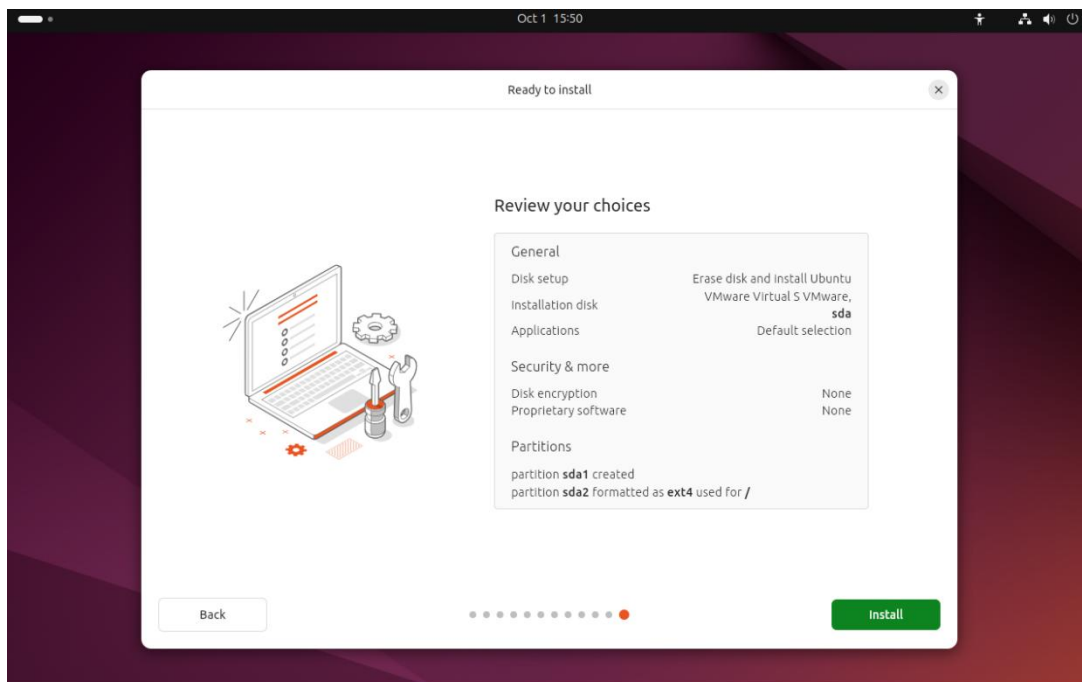
PART 1 : Install Ubuntu 20++

For Virtual Machine I using VMware Workstation.

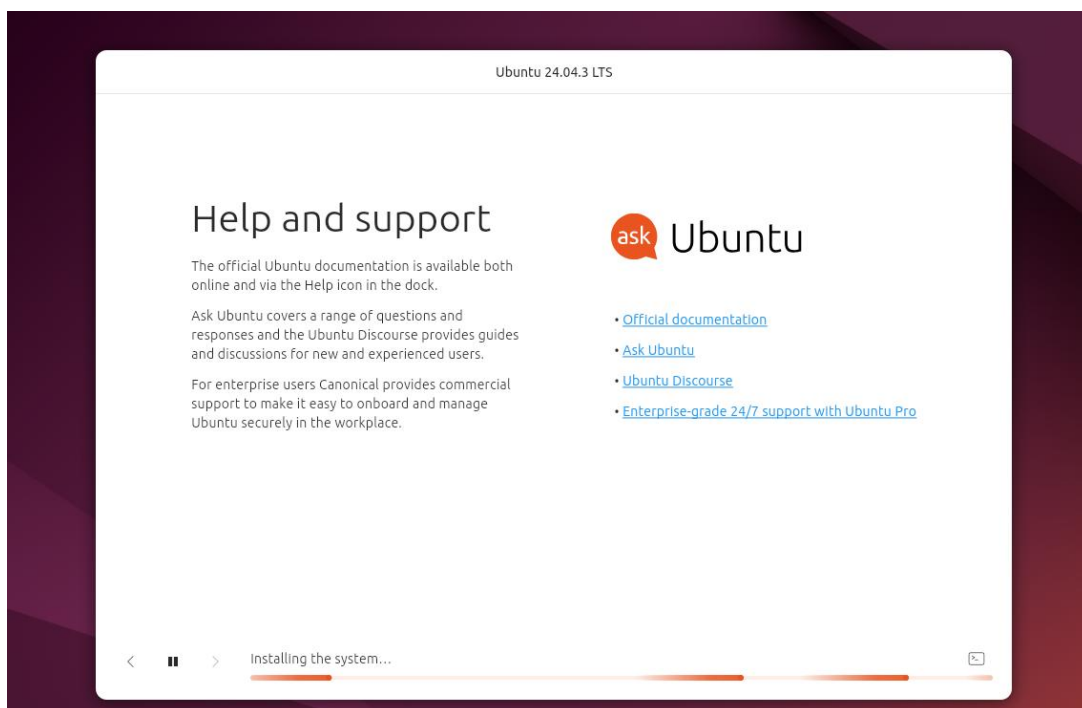
I download ubuntu desktop image version 24

 ubuntu-24.04.3-desktop-amd64	29/9/2025 9:48 AM	Disc Image File	6,197,156 KB
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There a few setup before we can install the ubuntu.



The ubuntu is installing the system.



Ubuntu is ready to use without an error

PART 2 : Setup nginx

I install nginx using command `sudo apt install nginx`

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ sudo apt install nginx
[sudo] password for ubuntu:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 64 not upgraded.
Need to get 564 kB of archives.
After this operation, 1,596 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

After installation complete, I use command `nginx -h` to see the option for use nginx

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ nginx -h
nginx version: nginx/1.24.0 (Ubuntu)
Usage: nginx [-?hvVtTq] [-s signal] [-p prefix]
              [-e filename] [-c filename] [-g directives]

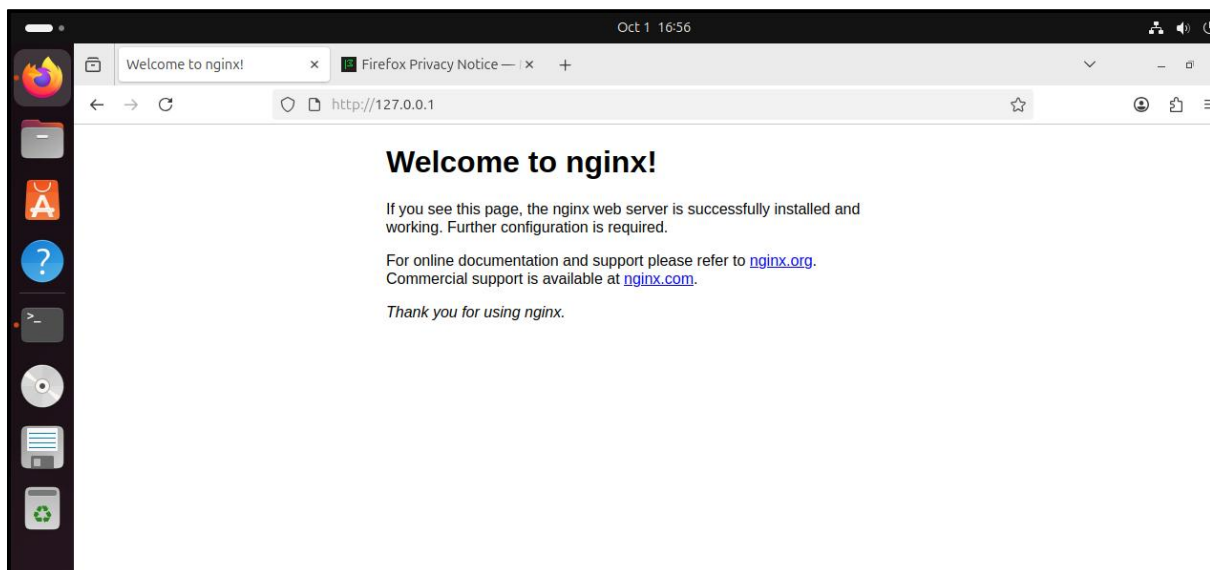
Options:
  -?,-h          : this help
  -v            : show version and exit
  -V            : show version and configure options then exit
  -t            : test configuration and exit
  -T            : test configuration, dump it and exit
  -q            : suppress non-error messages during configuration testing
  -s signal     : send signal to a master process: stop, quit, reopen, reload
  -p prefix     : set prefix path (default: /usr/share/nginx/)
  -e filename   : set error log file (default: stderr)
  -c filename   : set configuration file (default: /etc/nginx/nginx.conf)
  -g directives : set global directives out of configuration file
```

To see the activation is active or not with command `systemctl status nginx`

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: en>
   Active: active (running) since Wed 2025-10-01 16:31:01 +08; 13min ago
     Docs: man:nginx(8)
  Process: 4321 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_proce>
  Process: 4322 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (c>
 Main PID: 4352 (nginx)
    Tasks: 3 (limit: 4545)
   Memory: 2.4M (peak: 5.4M)
      CPU: 30ms
   CGroup: /system.slice/nginx.service
           └─4352 "nginx: master process /usr/sbin/nginx -g daemon on; master>
             └─4355 "nginx: worker process"
               └─4356 "nginx: worker process"

Oct 01 16:31:01 ubuntu-VMware-Virtual-Platform systemd[1]: Starting nginx.servi>
Oct 01 16:31:01 ubuntu-VMware-Virtual-Platform systemd[1]: Started nginx.servic>
lines 1-17/17 (END)
```

The test should see like this it mean nginx works.



PART 3 : Log File Overview

For this part I use `cat /var/log/syslog` to view system logs for ubuntu.

```
ubuntu@ubuntu-VMware-Virtual-Platform: ~  
ubuntu@ubuntu-VMware-Virtual-Platform:~$ sudo /var/log/nginx/access.log  
sudo: /var/log/nginx/access.log: command not found  
ubuntu@ubuntu-VMware-Virtual-Platform:~$ cat /var/log/syslog  
2025-10-01T08:29:30.163323+00:00 ubuntu-VMware-Virtual-Platform systemd-modules-  
load[400]: Inserted module 'lp'  
2025-10-01T08:29:30.163546+00:00 ubuntu-VMware-Virtual-Platform systemd-modules-  
load[400]: Inserted module 'ppdev'  
2025-10-01T08:29:30.163559+00:00 ubuntu-VMware-Virtual-Platform systemd-modules-  
load[400]: Inserted module 'parport_pc'  
2025-10-01T08:29:30.163563+00:00 ubuntu-VMware-Virtual-Platform systemd-modules-  
load[400]: Inserted module 'msr'  
2025-10-01T08:29:30.163566+00:00 ubuntu-VMware-Virtual-Platform systemd[1]: Fini  
shed systemd-tmpfiles-setup-dev-early.service - Create Static Device Nodes in /d  
ev gracefully.  
2025-10-01T08:29:30.163569+00:00 ubuntu-VMware-Virtual-Platform systemd[1]: Star  
ting systemd-journal-flush.service - Flush Journal to Persistent Storage...  
2025-10-01T08:29:30.163572+00:00 ubuntu-VMware-Virtual-Platform systemd[1]: Star  
ting systemd-sysusers.service - Create System Users...  
2025-10-01T08:29:30.163575+00:00 ubuntu-VMware-Virtual-Platform systemd-sysusers  
[438]: Creating group 'gamemode' with GID 986.  
2025-10-01T08:29:30.163583+00:00 ubuntu-VMware-Virtual-Platform systemd-sysusers  
[438]: Creating group 'gnome-initial-setup' with GID 985.  
2025-10-01T08:29:30.163586+00:00 ubuntu-VMware-Virtual-Platform systemd[1]: Fini  
shed systemd-sysusers.service - Create System Users.
```

This is the access log for the nginx

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ cat /var/log/nginx/access.log  
127.0.0.1 - - [01/Oct/2025:16:54:51 +0800] "GET / HTTP/1.1" 200 409 "-" "Mozilla  
/5.0 (X11; Ubuntu; Linux x86_64; rv:141.0) Gecko/20100101 Firefox/141.0"  
127.0.0.1 - - [01/Oct/2025:16:54:51 +0800] "GET /favicon.ico HTTP/1.1" 404 134 "  
http://127.0.0.1/" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:141.0) Gecko/2010  
0101 Firefox/141.0"
```


PART 4 : Monitoring Log files in Real Time

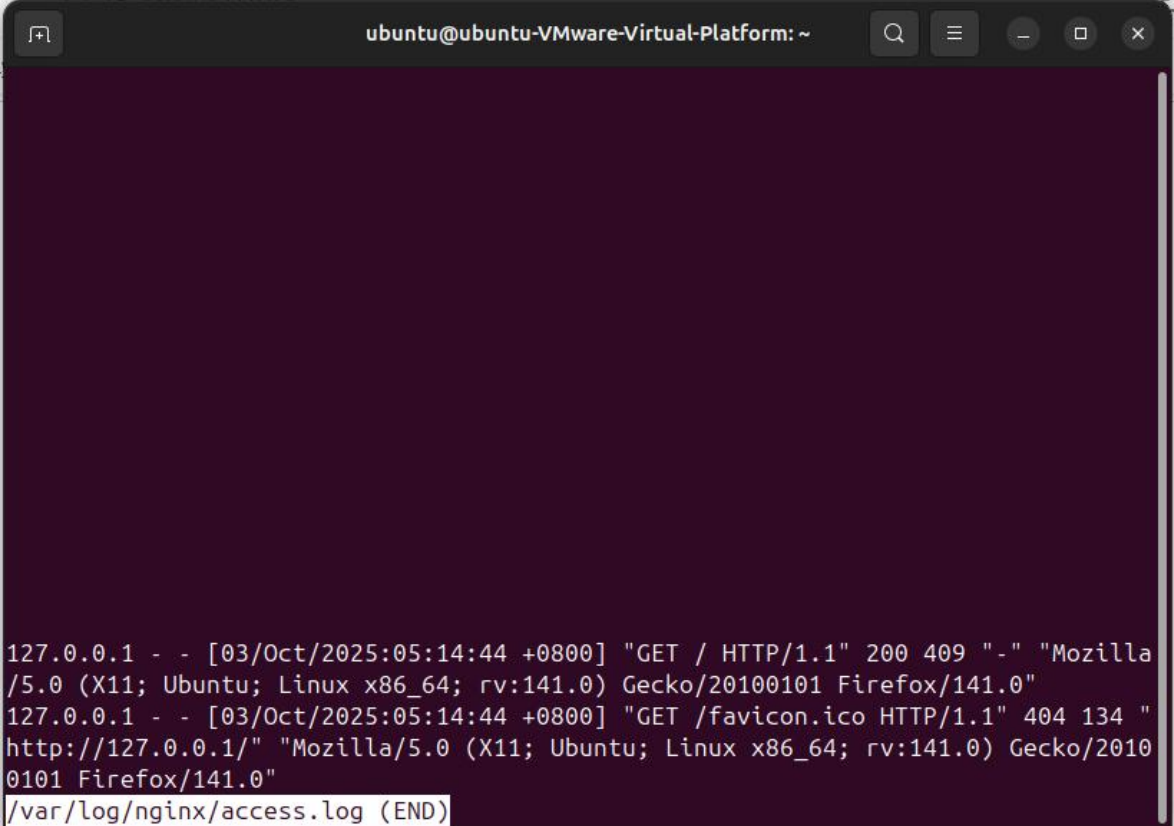
For this part I use 2 tools such as tail and less .

```
sudo tail /var/log/nginx/access.log .
```

tail is used to view the end of a file (newest logs)

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ sudo tail /var/log/nginx/access.log
127.0.0.1 - - [03/Oct/2025:05:14:44 +0800] "GET / HTTP/1.1" 200 409 "-" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:141.0) Gecko/20100101 Firefox/141.0"
127.0.0.1 - - [03/Oct/2025:05:14:44 +0800] "GET /favicon.ico HTTP/1.1" 404 134 "http://127.0.0.1/" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:141.0) Gecko/20100101 Firefox/141.0"
```

```
sudo less /var/log/nginx/access.log
```

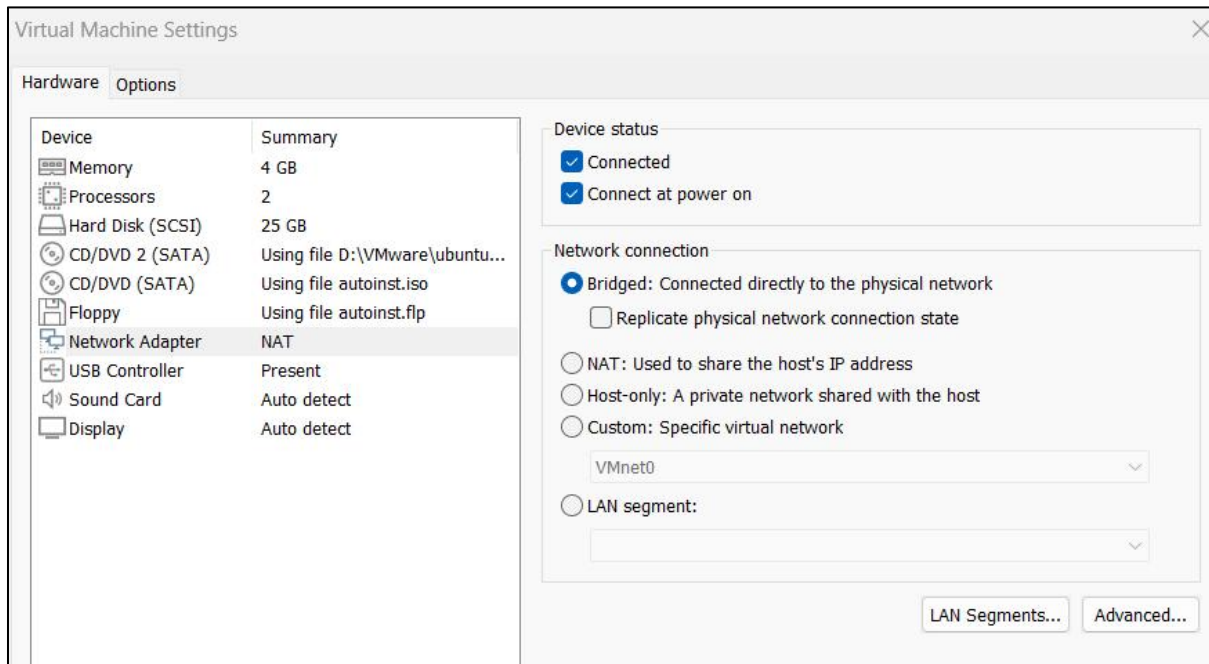


```
ubuntu@ubuntu-VMware-Virtual-Platform: ~
127.0.0.1 - - [03/Oct/2025:05:14:44 +0800] "GET / HTTP/1.1" 200 409 "-" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:141.0) Gecko/20100101 Firefox/141.0"
127.0.0.1 - - [03/Oct/2025:05:14:44 +0800] "GET /favicon.ico HTTP/1.1" 404 134 "http://127.0.0.1/" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:141.0) Gecko/20100101 Firefox/141.0"
/var/log/nginx/access.log (END)
```

As you can see, it displayed the same result for monitoring log files.

PART 5 :Review the Log

For this part, I changed the Ubuntu network adapter setting from NAT to Bridged.



Then I create a new folder and HTML file. After that, I add example for the html.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ sudo mkdir /var/www/html/RAIMI
ubuntu@ubuntu-VMware-Virtual-Platform:~$ sudo echo <h1>SAYA RAIMI</h1> | sudo tee /var/www/html/RAIMI/example.html
bash: syntax error near unexpected token `|'
ubuntu@ubuntu-VMware-Virtual-Platform:~$ echo <h1>SAYA RAIMI</h1> | sudo tee /var/www/html/RAIMI/example.html
bash: syntax error near unexpected token `|'
ubuntu@ubuntu-VMware-Virtual-Platform:~$ sudo echo <h1>SAYA RAIMI</h1> | tee /var/www/html/RAIMI/example.html
bash: syntax error near unexpected token `|'
ubuntu@ubuntu-VMware-Virtual-Platform:~$ sudo echo <h1>SAYA RAIMI</h1> tee /var/www/html/RAIMI/example.html
bash: h1: No such file or directory
ubuntu@ubuntu-VMware-Virtual-Platform:~$ echo <h1>SAYA RAIMI</h1> tee /var/www/html/RAIMI/example.html
bash: h1: No such file or directory
ubuntu@ubuntu-VMware-Virtual-Platform:~$ sudo echo <h1>RAIMI</h1> tee /var/www/html/RAIMI/example.html
bash: h1: No such file or directory
ubuntu@ubuntu-VMware-Virtual-Platform:~$ sudo echo "<h1>SAYA RAIMI</h1>" | sudo tee /var/www/html/RAIMI/example.html
<h1>SAYA RAIMI</h1>
```

mkdir is to create a new folder.

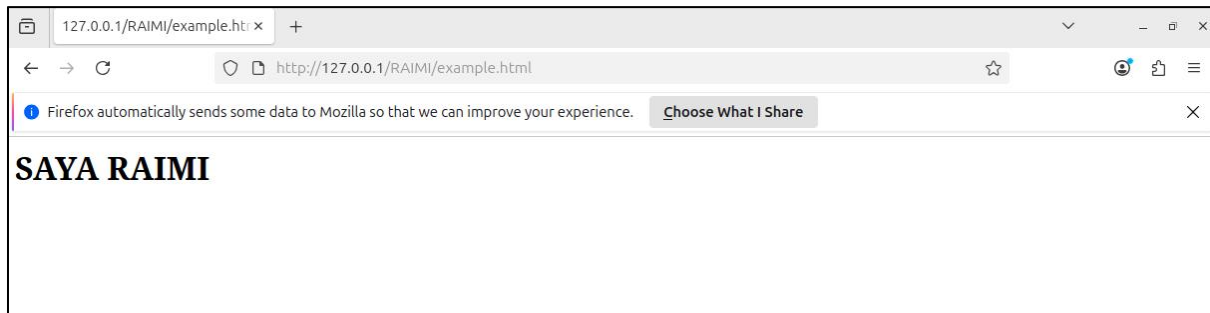
SAYA RAIMI is the output that will be displayed on the web.

tee is command to take output to both terminal and file.

After create the file , I want to test it on my friend laptop

Put command `http://127.0.0.1/RAIMI/example.html`

This is the output displayed and it successful to show my example.



And lastly I want to monitoring the log again

The monitoring is already done.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ sudo tail -f /var/log/nginx/access.log
127.0.0.1 - - [03/Oct/2025:05:14:44 +0800] "GET / HTTP/1.1" 200 409 "-" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:141.0) Gecko/20100101 Firefox/141.0"
127.0.0.1 - - [03/Oct/2025:05:14:44 +0800] "GET /favicon.ico HTTP/1.1" 404 134 "http://127.0.0.1/" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:141.0) Gecko/20100101 Firefox/141.0"
127.0.0.1 - - [03/Oct/2025:05:56:25 +0800] "GET /RAIMI/example.html HTTP/1.1" 200 49 "-" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:141.0) Gecko/20100101 Firefox/141.0"
```

Conclusion

This lab has provided hands-on experience in setting up a virtualized Linux environment and managing system log files using various tools. Through the installation of NGINX and real-time log monitoring with commands like tail, we gained practical knowledge on how system activities and user interactions are recorded. We also simulated a real-world scenario by attempting access to a server and reviewing the corresponding log entries. These skills are essential for identifying security incidents, ensuring system integrity, and maintaining operational transparency in network environments. Overall, this lab reinforces the importance of proper log management as a proactive measure in securing computer networks.

PRACTICAL TASK 1

CLO 1: Follow security procedures to prevent vulnerabilities and attacks in a network environment. (P3, PLO 3)

TOPIC: 2.4 Display Basic Linux Administration Task

NAME / REGISTRATION NO : MOHAMAD RAIMIHAKIM BIN MOHD KAHAR (25DDT23F1175)

CLASS : DDT5IS1

LECTURER : MOHD FAKHRU RAZIE BIN MUKHTAR

NO.	CRITERIA	3	2	1	WEIGHTAGE	MARKS
PRACTICAL TASK 1 – 100%						
1	Install Ubuntu 2+ VM	Students able to prepare ubuntu VM ISO successfully and independently without errors or problems.	Students able to prepare ubuntu VM ISO successfully and with a little guidance from the lecturer.	Students able to prepare ubuntu VM ISO successfully and with lecturer guidance .	10	(Score / 3) x 10 = _____
2	Setup nginx	Students able to install nginx successfully and independently without errors or problems.	Students able to install nginx successfully and with a little guidance from the lecturer.	Students able to install nginx successfully and with lecturer guidance.	20	(Score / 3) x20= _____

NO.	CRITERIA	3	2	1	WEIGHTAGE	MARKS
3	Log File Overview	Students able to show log file nginx conf successfully and independently without errors or problems.	Students able to show log file nginx conf successfully and with a little guidance from the lecturer.	Students able to show log file nginx conf successfully and with lecturer guidance.	20	(Score / 3) x 20= _____
4	Monitoring Log File in Real Time	Students able to using text-prestation tools successfully and independently without errors or problems.	Students able to using text-prestation tools successfully and with a little guidance from the lecturer.	Students able to using text-prestation tools successfully and with lecturer guidance.	20	(Score / 3) x 20= _____
5	Review the logs	Students able describe what happened successfully and independently without errors or problems.	Students able describe what happened successfully and with a little guidance from the lecturer.	Students able describe what happened successfully and with lecturer guidance.	30	(Score / 3) x 30 = _____
TOTAL						/100

Remarks: Zero (0) marks will be give if student do not answer / perform the criteria needed.